

Modular Arithmetic and $(\mathbb{Z}_{12}, +)$

Imagine you live in Clock World, where you can only do addition using the hour hand of a clock.



Figure 1: Clock World: A.K.A. Salvador Dali's *The Persistence of Memory*

You would be limited in how many numbers you had to work with. To see why, imagine that the hour hand is currently pointing to the number 11. Now advance by 15 hours. Would the hour hand be pointing to the number 26 (which is what $11 + 15$ is equal to)? What do you think? If you are unsure, look at the diagram below.

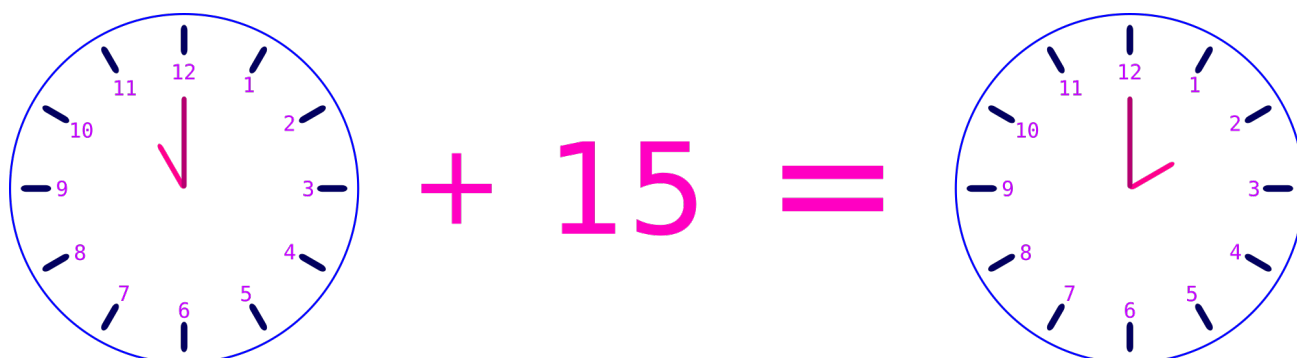
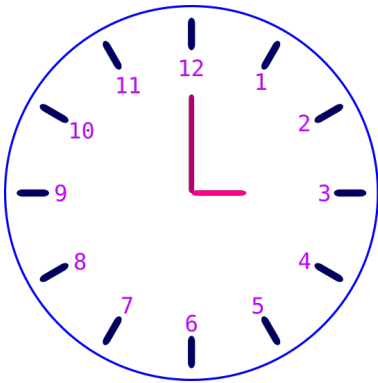
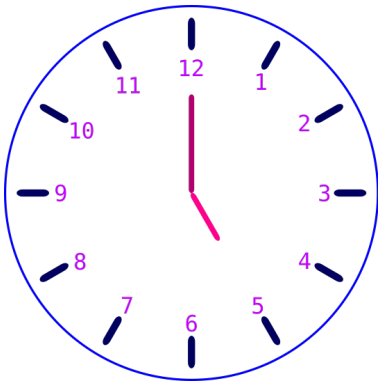


Figure 2: On a clock, $11 + 15 = 2$

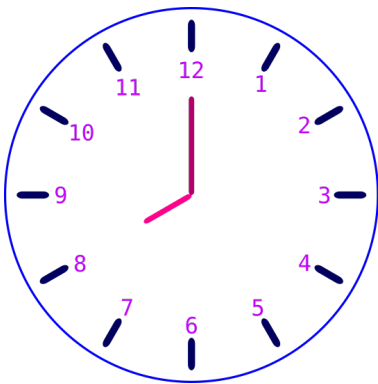
The number 26 isn't actually on a clock, so it can't be the number the hour hand is pointing to. Whenever the hour hand goes beyond 12, the count starts over again. If this makes sense to you, try to solve the following Clock World addition questions.



$$+ 17 =$$



$$+ 12 =$$



$$+ 26 =$$

You might think that using pictures of clocks is a cumbersome way of doing mathematics. I agree, and so does every sane mathematician on earth. That's why we have mathematical notation. Allow me to introduce you to the mathematical object $(\mathbb{Z}_{12}, +)$. This mathematical object is technically a **group**, but we will get into what groups are at a later point. Let's examine the components of this object.

The notation $\mathbb{Z}_{12}$ represents the set of 'integers' from 0 to 11 (technically, they are equivalence classes, not integers, but don't worry about that yet). To make things clear for you, you can think of $\mathbb{Z}_{12}$ as

$$\mathbb{Z}_{12} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}.$$

The only difference between the numbers in $\mathbb{Z}_{12}$ and the numbers on a clock is that $\mathbb{Z}_{12}$ has a 0 where a clock has 12. But, in Clock World, 12 and 0 are equal, so this isn't a huge problem. Here's a new clock, just to make this a little clearer.

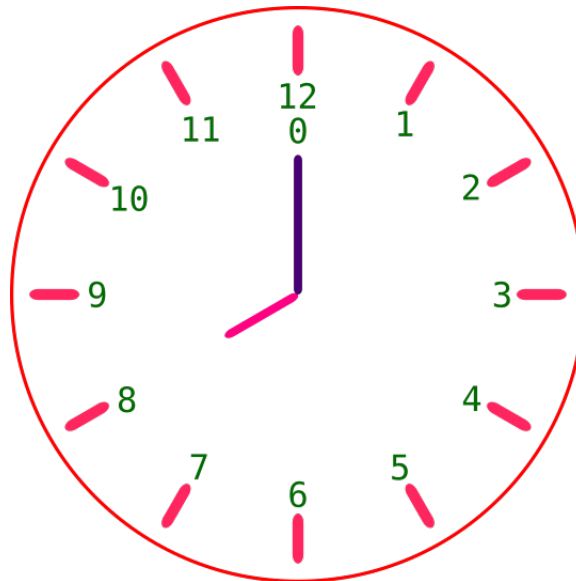


Figure 3: In Clock World, $12 = 0$

If you are familiar with military time, then you know that 12AM is equivalent to 00:00. This might help your understanding. You might also know that 1PM is equivalent to 13:00 and 2PM is equivalent to 14:00. Likewise, in $\mathbb{Z}_{12}$, since we only have the integers 0 to 11 to work with, $13 = 1$ and $14 = 2$. What would 17 be equal to?

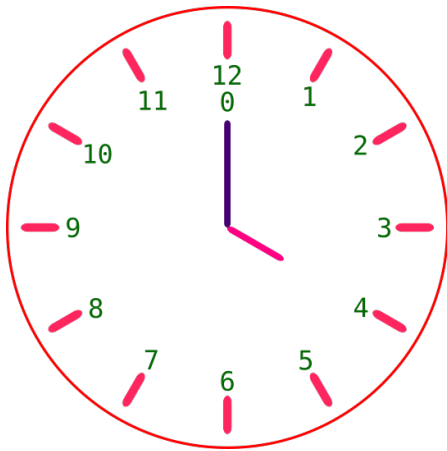
As for the $+$ in $(\mathbb{Z}_{12}, +)$, it represents something called 'addition modulo 12.' You can think of it as 'wrap-around' addition, i.e., addition such that you can never get a sum higher than 11. If you added two numbers together and got 12, you would wrap-around to 0; if you added two numbers together and got 13, you would wrap-around to 1; etc.

So the mathematical object $(\mathbb{Z}_{12}, +)$ is a model of a clock whose hour hand can be moved discretely in a clockwise direction. Here, 'discretely' means that the hour hand can't land on 8.5 or 11.5678; it can only land on integers. In other words, we aren't moving the hour hand through a continuum, if this word means anything to you. If it doesn't, don't worry about it.

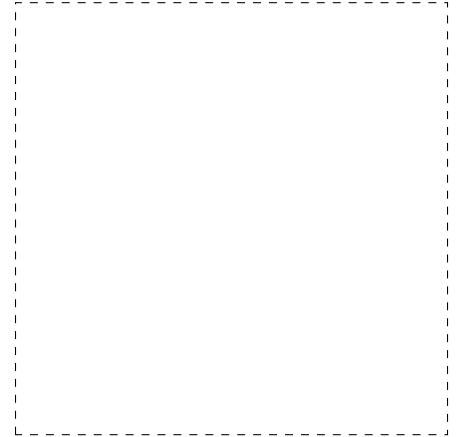
If I wanted to ask you to perform a Clock World addition question without using clock pictures, I would write something like this:

Find the sum $4 + 25 \pmod{12}$.

This question is equivalent to the following picture question:



+ 25 =



Exercises

Instructions: Compute the following sums modulo 12.

(1.) $2 + 13 \pmod{12} =$

(2.) $9 + 24 \pmod{12} =$

(3.) $0 + 36 \pmod{12} =$

(4.) $7 + 19 \pmod{12} =$

(5.) $1 + 27 \pmod{12} =$